Paul S. Lou

Postdoctoral Researcher | Bocconi University

paul.lou@unibocconi.it | paullou.me | Google Scholar | GitHub

Interests

AI safety; LLM reasoning; inference; theoretical and applied cryptography: public-key encryption, zero-knowledge proof systems, new hardness assumptions; information theory; quantum algorithms.

Research Experience

Bocconi University *Sep. 2025–present*
Postdoctoral Researcher with Dr. Alon Rosen.

NTT Research *Sep.–Dec. 2023*
Research Intern with Dr. Abhishek Jain.

Carnegie Mellon University *Oct. 2022–Jan. 2023*
Visiting Scholar with Dr. Aayush Jain.

Simons Institute for the Theory of Computing *May–Jun. 2022*
Visiting Graduate Student for summer cluster: "Lattices and Beyond."

University of Pennsylvania *May 2016–Dec. 2018*
Undergraduate Researcher with Dr. Nadia Heninger.

Education

University of California, Los Angeles *2019–May 2025*
Doctor of Philosophy in Computer Science.
Advised by Dr. Amit Sahai.

University of Pennsylvania *Dec. 2018*
Management & Technology Dual Degree Program:
B.S.E. in Mathematics and Computer Science. School of Engineering and Applied Sciences.
Advised by Dr. Nadia Heninger.
B.S. in Economics, concentration in Statistics. The Wharton School.

*All authors listed by alphabetical order, as is usual in cryptographic research.

In Submission*

S3 **Worst-Case Hardness for Succinct and Decomposed Learning with Errors**
Damiano Abram, Gal Arnon, Valerio Cini, **Paul Lou**, Giulio Malavolta, Lawrence Roy.
2026 | https://eprint.iacr.org/2026/2443
Subsumes Decomposed LWE is Equivalent to Succinct LWE.

S2 **Limitations on Autoregressive Token Generation via Sequential Sampling**
Gal Arnon, Andrej Bogdanov, **Paul Lou**, Giulio Malavolta, Alon Rosen, Lorenzo Taschin.
2026 | https://arxiv.org/abs/2610.14615

S1 **Post-Quantum PKE and More from a Noisy Unstructured Linear Algebraic Assumption: Beyond LWE and LPN**
Riddhi Ghosal, **Paul Lou**, Amit Sahai.
2026 | https://eprint.iacr.org/2026/2113

Publications*

P12 **Expanders Meet Reed–Muller: Easy Instances of Noisy k-XOR**
Jarosław Błasiok, **Paul Lou**, Alon Rosen, Madhu Sudan.
FOCS 2026 [CORE A*] | https://arxiv.org/abs/2604.04188

P11 **Quantum Advantage via Solving Multivariate Polynomials**
Pierre Briaud, Itai Dinur, Riddhi Ghosal, Aayush Jain, **Paul Lou**, Amit Sahai.
SODA 2026 [CORE A*] | https://arxiv.org/abs/2509.07276

P10 **Fully Anonymous Secret Sharing**
Allison Bishop, Matthew Green, Yuval Ishai, Abhishek Jain, **Paul Lou**.
Crypto 2025 [CORE A*] | https://eprint.iacr.org/2025/1984

P9 **Post-Quantum PKE from Unstructured Noisy Linear Algebraic Assumptions: Beyond LWE and Alekhnovich's LPN**
Riddhi Ghosal, Aayush Jain, **Paul Lou**, Amit Sahai, Neekon Vafa.
Eurocrypt 2025 [CORE A*] | https://eprint.iacr.org/2025/844

P8 **Witness Semantic Security**
Paul Lou, Nathan Manohar, Amit Sahai.
Eurocrypt 2024 [CORE A*] | https://eprint.iacr.org/2024/1518

P7 **Computational Wiretap Coding from Indistinguishability Obfuscation**
Yuval Ishai, Aayush Jain, **Paul Lou**, Amit Sahai, Mark Zhandry.
Crypto 2023 [CORE A*] | https://eprint.iacr.org/2023/1270

P6 **Hard Languages in NP ∩ coNP and NIZK Proofs from Unstructured Hardness**
Riddhi Ghosal, Yuval Ishai, Alexis Korb, Eyal Kushilevitz, **Paul Lou**, Amit Sahai.
STOC 2023 [CORE A*] | https://dl.acm.org/doi/10.1145/3564246.3585119

P5 **Polynomial-Time Cryptanalysis of the Subspace Flooding Assumption for Post-Quantum *iO***
Aayush Jain, Rachel Lin, **Paul Lou**, Amit Sahai.
Eurocrypt 2023 [CORE A*] | https://eprint.iacr.org/2022/1637

P4 **Efficient NIZKs from LWE via Polynomial Reconstruction and "MPC in the Head"**
Riddhi Ghosal, **Paul Lou**, Amit Sahai.
Asiacrypt 2022 [CORE A] | https://eprint.iacr.org/2022/370

P3 **Beyond the Csiszár-Korner Bound: Best-Possible Wiretap Coding via Obfuscation**
Yuval Ishai, Alexis Korb, **Paul Lou**, Amit Sahai.
Crypto 2022 [CORE A*] | https://eprint.iacr.org/2022/343
Invited and accepted submission to *Journal of Cryptology*.

P2 **Relinearization Attack on LPN over Large Fields**
Paul Lou, Amit Sahai, Varun Sivashankar.
CFAIL 2022 [CORE: not listed] | https://doi.org/10.1093/comjnl/bxad070
Crypto-affiliated workshop. Invited and accepted submission to a special edition of *The Computer Journal*.

P1 **Post-quantum RSA**
Daniel J. Bernstein, Nadia Heninger, **Paul Lou**, Luke Valenta.
PQCrypto 2017 [CORE C] | https://eprint.iacr.org/2017/351

Preprints

W2 **A Note on the Pseudorandomness of Low-Degree Polynomials over the Integers**
Aayush Jain, Alexis Korb, **Paul Lou**, Amit Sahai.
2021 | https://eprint.iacr.org/2021/1415

W1 **Expanding COVID-19 Symptom Screening to Retail, Restaurants, and Schools by Preserving Privacy Using Relaxed Digital Signatures**
Brandon Jew, Alexis Korb, **Paul Lou**, Jeffrey N. Chiang, Ulzee An, Amit Sahai, Eran Halperin, Eleazar Eskin.
2020 | https://www.medrxiv.org/content/10.1101/2020.08.06.20169839v2

Talks

- Limitations on Autoregressive Token Generation via Sequential Sampling
  Bocconi seminar, Oct. 2026
- Quantum Advantage via Solving Multivariate Polynomials
  University of Texas, Austin, Dec. 2025

· Fully Anonymous Secret Sharing
  Crypto 2025
  Stanford, Sep. 2025
· Ph.D. Thesis Defense
  University of California, Los Angeles, May 2025
· Witness Semantic Security
  Eurocrypt 2024
· Computational Wiretap Coding from Indistinguishability Obfuscation
  Simons Institute Summer 2025
  Conference on Information-Theoretic Cryptography 2024
  Crypto 2023
· Hard Languages in NP ∩ coNP and NIZK Proofs from Unstructured Hardness
  Carnegie Mellon University, Apr. 2024
· Beyond the Csiszár-Korner Bound: Best-Possible Wiretap Coding via Obfuscation
  University of California, Los Angeles, Oral Qualifying Exam, Nov. 2023
· Polynomial-Time Cryptanalysis of the Subspace Flooding Assumption for Post-Quantum *iO*
  Eurocrypt 2023
· Efficient NIZKs from LWE via Polynomial Reconstruction and "MPC in the Head"
  Asiacrypt 2022
· Relinearization Attack on LPN over Large Fields
  CFAIL 2022

Academic Service

Program committee (PC) member for Crypto 2027, Asiacrypt 2026, CCS 2026, Oakland 2023 Posters Program.
Reviewer for *Journal of Cryptology* and *Designs, Codes and Cryptography*.
External reviewer for Crypto ('22, '25, '26), Eurocrypt ('24, '25, '26), STOC ('24, '25, '26), FOCS ('26), TCC ('23, '24, '25), IEEE S&P ('24), Asiacrypt ('24), ITCS ('27), ICLR ('27).

Teaching

Instructor
(Co-lead Instructor) CS-289: Advanced Topics in Cryptography (Quantum Cryptography), Spring 2023, University of California, Los Angeles.

Teaching assistant
CS-181: Formal Languages and Automata Theory, Winter 2021, Winter 2022, Winter 2024, University of California, Los Angeles.
CIS-556: Cryptography (Graduate-level), Fall 2018, UPenn.
CIS-548: Operating Systems (Graduate-level), Spring 2018, UPenn.
CIS-380: Operating Systems, Fall 2017, Fall 2018 (Head TA), UPenn.
CIS-262: Theory of computation: Automata, Computability, & Complexity, Fall 2016, UPenn.

Mentorship
Varun Sivashankar (UCLA Undergraduate) → Princeton Ph.D. in Applied Mathematics.

Programming Languages

Preferred	Python, C, C++
Experienced	OCaml, Java, Sage, Magma

Personal Information

Languages	English (Mothertongue); Mandarin (Bilingual); French (B2); German (B1); Italian (B1).
Nationality	US Citizenship
Email	paul.lou@unibocconi.it
Misc. Interests	Skiing · Swimming · Hot Chocolate